

Quitters: a correction to Proposition 20 and an answer to Question 26

On S. A. Alexander, *Intelligence via Ultrafilters* (2019)

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Status. Elementary hand proofs, with an exact-arithmetic check script. No independent human review. A bounded search did not locate a published answer; priority is not established.

Source. S. A. Alexander, *Intelligence via ultrafilters: structural properties of some intelligence comparators of deterministic Legg-Hutter agents*, Journal of Artificial General Intelligence 10(1), 2019. [arXiv:1910.09721](https://arxiv.org/abs/1910.09721), [doi:10.2478/jagi-2019-0003](https://doi.org/10.2478/jagi-2019-0003).

Summary.

1. **Proposition 20 is false as printed for every $r \leq 0$, and true for every $r > 0$.** The initial reward arrives before any action, so neither the skipping hypothesis nor the bounded-reward hypothesis constrains it. It alone can reach $r + 1$.
2. **Question 26, under Legg and Hutter’s own normalisation** (rewards in $[0, 1]$, total reward at most 1): the answer is **yes for every $r \geq 0$ and no for every $r < 0$** . The negative cases turn on the same initial-reward edge.

Check script: `quitters_check.py` in this folder, which writes `quitters_check.json`.

The setting (the paper’s Definitions 1, 17-19)

- The environment emits an **initial** reward-observation pair $(r_1, o_1) = e(\cdot)$ *before* the agent’s first action. A deterministic agent (DLHA) maps reward-observation histories to actions.
- An environment **respects skipping** if the reward in response to the action “skip” (0) is 0. It has **bounded rewards** if every reward is at most 1. Rewards may be negative.
- The **quitter** A_r acts as A while $r_1 + \dots + r_n < r$, and skips forever once the sum is at least r .

Proposition 20 (as printed). Let E be an electorate with bounded rewards that respects skipping, let $A >_E B$, and let $r \in \mathbb{R}$. If $B_r =_E B$, then $A_{r+1} >_E B$.

Question 26. Let Γ' be Legg-Hutter universal intelligence restricted to environments that respect skipping and have bounded rewards. If $\Gamma'(A) > \Gamma'(B)$ and $\Gamma'(B_r) = \Gamma'(B)$, must $\Gamma'(A_{r+1}) > \Gamma'(B)$?

1. Proposition 20 fails for every $r \leq 0$ and holds for $r > 0$

The gap. The proof asserts that B_r ’s total is below $r + 1$. The argument is that B_r stops at the first n with $r_1 + \dots + r_n \geq r$, and the last reward is at most 1. That is right when $n \geq 2$, since the previous partial sum is below r .

When $n = 1$, the total is just r_1 , the initial reward. That reward is *not* a response to a skip, so neither hypothesis bounds it below 1. If $r_1 \geq r + 1$, which is possible exactly when $r \leq 0$, the step fails.

Counterexample (any $r \leq 0$). Let every e_n be the same environment e . It gives initial reward 1, then reward $\frac{1}{2}$ in response to a first action that is not “skip”, and 0 otherwise.

- It has bounded rewards and respects skipping. Definition 6 allows any sequence of environments, including constant sequences.
- Let A = always act 1, with total $3/2$, and B = always skip, with total 1. Then $A >_E B$.
- $B_r = B$, since B only skips anyway, so $B_r =_E B$.
- A_{r+1} sees $r_1 = 1 \geq r + 1$ and skips at once, with total 1. So $A_{r+1} =_E B$, **not** $A_{r+1} >_E B$.

The script checks this at $r = -1, -\frac{1}{2}$ and 0. At $r = \frac{1}{4}$ the conclusion holds.

Repair. For $r > 0$ the case $n = 1$ is harmless: $r_1 \geq r > 0$ and $r_1 \leq 1$ give $r_1 < r + 1$. The printed proof then goes through unchanged. So Proposition 20 holds with the hypothesis $r > 0$. For $r \leq 0$ it needs an extra condition, for instance that initial rewards are below $r + 1$.

2. Question 26 under Legg-Hutter normalisation: yes iff $r \geq 0$

Γ is Legg and Hutter’s measure. Its environments have rewards in $[0, 1]$ and **total reward at most 1** for every agent, and every environment of the restricted class gets a positive weight $2^{-(K)}$.

Two facts used throughout.

1. **Quitting never helps when rewards are non-negative.** Couple the two runs on the same environment randomness. B_r earns B ’s rewards until it stops, and 0 afterwards; B earns ≥ 0 afterwards. So $V_e(B_r) \leq V_e(B)$ in every environment. All weights are positive, so $\Gamma'(B_r) = \Gamma'(B)$ forces $V_e(B_r) = V_e(B)$ for every e . The same argument gives $V_e(\text{Skip}) \leq V_e(X)$ for every agent X .
2. **Partial sums never exceed 1**, since totals are at most 1.

$r > 0$: yes. The stopping condition, a partial sum $\geq r + 1 > 1$, never occurs. So A_{r+1} plays exactly as A on every reachable history, and $\Gamma'(A_{r+1}) = \Gamma'(A) > \Gamma'(B)$.

$r = 0$: yes. A_1 stops only once its partial sum reaches 1. The total is at most 1, so every later reward is 0, whichever agent plays. So $V_e(A_1) = V_e(A)$, and $\Gamma'(A_1) = \Gamma'(A) > \Gamma'(B)$.

$r < 0$: no. Put $c = r + 1 < 1$ and let $B = \text{Skip}$. Every partial sum is $\geq 0 > r$, so $B_r = \text{Skip} = B$ and the hypothesis holds.

- **If $c \leq 0$** , take A = always act 1.
 - A_{r+1} skips from the start, so $\Gamma'(A_{r+1}) = \Gamma'(\text{Skip}) = \Gamma'(B)$.
 - Yet $\Gamma'(A) > \Gamma'(\text{Skip})$: the class contains an environment that rewards acting, and A never scores below Skip .
- **If $0 < c < 1$** , take A = “act 1 if $r_1 \geq c$, else skip”.
 - *The gain is real.* The class contains e^* , with initial reward $q \in [c, 1]$ (rational) and then $(1 - q)/2$ for acting once. A gains there, so $\Gamma'(A) > \Gamma'(B)$.
 - *The quitter throws it away.* A_c skips at once whenever $r_1 \geq c$. When $r_1 < c$ it copies A , which skips, so its partial sum never reaches c . On every trajectory A_{r+1} scores exactly as Skip , so $\Gamma'(A_{r+1}) = \Gamma'(B)$.

A randomized check found no violation in 90 cases with $r \geq 0$, and a counterexample in all 110 cases with $r < 0$. It uses positive stand-in weights, since the arguments above do not depend on the weights’ values.

Reading.

- Under Legg and Hutter’s own normalisation, Question 26 is nearly degenerate for $r > 0$: a total ≤ 1 environment never lets a quitter with threshold above 1 quit.
- Its content sits at $r \leq 0$, and it is decided by the **same initial-reward edge case** that breaks Proposition 20.
- In the ultrafilter setting totals can exceed 1. That is why $r = 0$ fails there but holds here.

Scope

- The answer takes Γ to be Legg-Hutter 2007, with rewards in $[0, 1]$ and totals at most 1.
- The paper adds “bounded rewards” to Γ' as if Γ lacked it. That suggests it may have intended a class with larger totals, or with negative rewards. Under such a normalisation:
 - quitting can *help*, by avoiding later punishment, so the per-environment equality in fact 1 is lost;
 - the averaged hypothesis $\Gamma'(B_r) = \Gamma'(B)$ then allows cancellations;
 - a counterexample for $r > 0$ would need explicit control of universal weights.

That variant is **not** settled here. A conjecture about it is in 09-further-results.

- Questions 27-28 (the Hernández-Orallo-Dowe and Hibbard analogues) are untouched.
- Both results are elementary. The correction concerns the printed statement; the intended use ($r > 0$) is unaffected.

References

- S. Legg and M. Hutter, *Universal intelligence: a definition of machine intelligence*, Minds and Machines 17(4), 2007, 391-444.
- S. A. Alexander, *Intelligence via ultrafilters*, as above: Definitions 1, 6, 17-19, Proposition 20, Question 26.